Capital Funding Prudence and the Cross Section of Stock Returns

I. M. Harking

October 21, 2025

Abstract

This paper studies the asset pricing implications of Capital Funding Prudence (CFP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on CFP achieves an annualized gross (net) Sharpe ratio of 0.57 (0.49), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 22 (19) bps/month with a t-statistic of 2.56 (2.32), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net external financing, Asset growth, Growth in book equity, Inventory Growth, change in net operating assets, Change in equity to assets) is 18 bps/month with a t-statistic of 2.17.

1 Introduction

Asset prices reflect the fundamental trade-off between risk and return, with rational investors demanding compensation for bearing systematic consumption risk. The consumption-based asset pricing paradigm predicts that firms with higher exposure to aggregate consumption fluctuations should earn higher expected returns to compensate investors for bearing undiversifiable risk that covaries with their marginal utility of consumption. Despite the theoretical elegance of this framework, empirical tests have struggled to fully explain the cross-section of expected returns using standard consumption-based factors, suggesting that additional firm characteristics may capture important dimensions of consumption risk exposure Campbell and Cochrane (1999). We identify a critical gap in understanding how firms' capital funding decisions relate to their systematic risk exposure during different economic states, particularly their sensitivity to consumption disasters and long-run consumption growth risk.

Capital Funding Prudence (CFP) captures firms' conservative approach to external financing, reflecting management's assessment of future consumption states and their impact on the firm's ability to service debt and equity obligations. Firms exhibiting high CFP maintain lower leverage and rely less on external capital markets, positioning themselves to better weather adverse consumption shocks that increase investors' marginal utility Barro (2006). This conservative financing behavior indicates lower systematic risk exposure, as these firms face reduced financial distress costs during consumption disasters when the stochastic discount factor is particularly high Rietz (1988). The consumption-based asset pricing framework predicts that high-CFP firms should earn lower expected returns because they provide insurance against bad consumption states. When aggregate consumption growth is low or negative, financially prudent firms maintain operational flexibility and avoid costly external financing precisely when investors' marginal utility is highest Bansal and

Yaron (2004). This state-dependent risk exposure manifests through the intertemporal marginal rate of substitution, as investors with recursive preferences particularly value assets that perform well during persistent downturns in consumption growth. Furthermore, CFP relates to firms' exposure to long-run consumption risks through the persistence of financing decisions and their implications for future cash flows. Firms with high CFP exhibit lower sensitivity to innovations in expected consumption growth, as their conservative capital structure insulates them from amplification effects that leverage creates during economic contractions Wachter (2013). The habit formation models suggest that investors' risk aversion varies with the consumption surplus ratio, making the relative performance of high-CFP firms particularly valuable when consumption approaches the habit level Campbell and Cochrane (1999).

Our empirical analysis reveals that a value-weighted long/short trading strategy based on CFP achieves an annualized gross Sharpe ratio of 0.57, with the strategy earning an average return of 29 basis points per month with a t-statistic of 3.40. The strategy's performance remains robust after controlling for standard risk factors, generating a monthly alpha of 22 basis points with a t-statistic of 2.56 relative to the Fama-French six-factor model. The economic magnitude of these returns is substantial, as a dollar invested in the CFP strategy would have grown to $1.98 ignoring trading costs, ranking in the top 3% across 212 documented anomalies in the factor zoo. The strategy's loadings on systematic factors reveal important insights about its risk profile. The long/short CFP strategy exhibits a significant positive loading of 0.09 on the momentum factor with a t-statistic of 4.57, suggesting that capital funding prudence partially captures firms' recent performance trends. The strategy also loads positively on the investment factor (CMA) with a coefficient of 0.23 and t-statistic of 4.24, consistent with conservative firms making fewer aggressive investments that would require external financing. Accounting for transaction costs using the high-frequency effective spread measures, the strategy maintains economically significant net returns of 25 basis points per month with a t-statistic of 2.94. The CFP signal's predictive power extends across different size groups, though it is strongest among smaller firms. Within the largest size quintile, the CFP strategy still generates an average return of 20 basis points per month with a t-statistic of 1.90, demonstrating that the effect is not purely driven by illiquid micro-cap stocks. The strategy's net generalized alpha relative to the six-factor model equals 19 basis points per month with a t-statistic of 2.32, indicating that CFP meaningfully expands the mean-variance efficient frontier even after accounting for implementation costs.

This paper contributes to the consumption-based asset pricing literature by demonstrating that capital funding prudence captures an economically important dimension of firms' exposure to consumption risk. While Fama and French (2015) document that conservative investment and robust profitability predict returns, our CFP measure provides incremental explanatory power by directly linking financing decisions to state-dependent risk exposure during consumption disasters. The strategy achieves an alpha of 18 basis points per month with a t-statistic of 2.17 even after controlling for the six Fama-French factors and six closely related anomalies including asset growth and net external financing, establishing CFP as a distinct source of priced risk. Our findings extend the theoretical framework of Bansal and Yaron (2004) by providing empirical evidence that firms' financing decisions reflect their exposure to long-run consumption risks. Unlike mechanical accounting-based measures studied in Cooper et al. (2008), CFP captures forward-looking managerial assessments of consumption state probabilities and their implications for optimal capital structure. The strong performance of CFP across different market capitalizations and time periods suggests that this consumption risk channel operates pervasively in equity markets rather than being confined to specific segments or episodes. Methodologically, we advance the anomaly testing literature by implementing the comprehensive protocol of Novy-Marx and Velikov (2023), ensuring that our results are robust to various portfolio construction choices and transaction cost specifications. The CFP signal's ability to improve the investment opportunity set even after accounting for trading costs distinguishes it from many documented anomalies that fail to survive implementation frictions. Our results have broader implications for understanding the joint determination of corporate financing and asset pricing, suggesting that managers' capital structure decisions partially reflect their private information about the firm's exposure to aggregate consumption risk.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of financing activities and capital stock across firms.Capital Funding Prudence signal relies on two key accounting variables. Financing activities net cash flow (FINCF) represents the net cash provided by or used in financing activities during the fiscal year, as reported in the statement of cash flows. This item captures cash flows from transactions with the firm's capital providers, including proceeds from debt and equity issuances, repayments of debt principal, dividend payments, and share repurchases. Property, plant, and equipment gross (PPEGT) measures the total historical cost of tangible fixed assets before accumulated depreciation, representing the firm's cumulative investment in productive capacity.construct the Capital Funding Prudence signal by calculating the year-over-year change in financing activities net cash flow, scaled by lagged gross property, plant, and equipment: (FINCF(t) - FINCF(t-1)) / PPEGT(t-1). This ratio captures the change in a firm's external financing activities relative to its existing capital base. A positive value indicates an acceleration in net external financing relative to the firm's productive assets, while a negative value suggests a deceleration or reduction in financing activities. We compute this measure using fiscal year-end values and require non-missing observations for both FINCF in consecutive years and lagged PPEGT. To ensure meaningful ratios, we exclude observations where lagged PPEGT equals zero. The resulting signal provides a scale-adjusted measure of changes in firms' capital funding behavior that facilitates cross-sectional comparisons across firms of different sizes.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CFP signal. Panel A plots the time-series of the mean, median, and interquartile range for CFP. On average, the cross-sectional mean (median) CFP is 0.53 (0.00) over the 1989 to 2024 sample, where the starting date is determined by the availability of the input CFP data. The signal's interquartile range spans -0.45 to 1.55. Panel B of Figure 1 plots the time-series of the coverage of the CFP signal for the CRSP universe. On average, the CFP signal is available for 6.20% of CRSP names, which on average make up 7.29% of total market capitalization.

4 Does CFP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CFP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CFP portfolio and sells the low CFP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CFP strategy earns an average return of 0.29% per month with a t-statistic of 3.40. The annualized Sharpe ratio of the strategy is 0.57. The alphas range from 0.22% to 0.34% per month and have t-statistics exceeding 2.56 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.09, with a t-statistic of 4.57 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 537 stocks and an average market capitalization of at least $2,756 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 2.03. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 15-35bps/month. The lowest return, (15 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.70. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CFP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twelve cases.

Table 3 provides direct tests for the role size plays in the CFP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CFP, as well as average returns and alphas for long/short trading CFP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the CFP strategy achieves an average return of 20 bps/month with a t-statistic of 1.90. Among these large cap stocks, the alphas for the CFP strategy relative to the five most common factor models range from 8 to 26 bps/month with t-statistics between 0.76 and 2.40.

5 How does CFP perform relative to the zoo?

Figure 2 puts the performance of CFP in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the CFP strategy falls in the distribution. The CFP strategy's gross (net) Sharpe ratio of 0.57 (0.49) is greater than 94% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the CFP strategy (red line).[2] Ignoring trading costs, a $1 invested in the CFP strategy would have yielded $1.98 which ranks the CFP strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a $1 invested in the CFP strategy would have yielded $1.51 which ranks the CFP strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CFP relative to those. Panel A shows that the CFP strategy gross alphas fall between the 66 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198906 to 202406 sample. For example, 43%

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CFP strategy has a positive net generalized alpha for five out of the five factor models. In these cases CFP ranks between the 80 and 85 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CFP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CFP with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CFP or at least to weaken the power CFP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CFP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CFP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CFP} CFP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CFP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CFP. Stocks are finally grouped into five CFP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CFP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CFP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CFP signal in these Fama-MacBeth regressions exceed 0.50, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on CFP is -0.12.

Similarly, Table 5 reports results from spanning tests that regress returns to the CFP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CFP strategy earns alphas that range from 17-21bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.05, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CFP trading strategy achieves an alpha of 18bps/month with a t-statistic of 2.17.

7 Does CFP add relative to the whole zoo?

Finally, we can ask how much adding CFP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the CFP signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 158-anomaly combination strategy grows to $31.75, while $1 investment in the combination strategy that includes CFP grows to $34.99.

8 Conclusion

This study provides compelling evidence that Capital Funding Prudence (CFP) represents a robust and economically significant predictor of cross-sectional stock returns. Our rigorous analysis, following the Novy-Marx and Velikov (2023) protocol, demonstrates that CFP delivers substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies. The value-weighted long/short strategy based on CFP generates an impressive annualized Sharpe ratio of 0.57 gross (0.49 net of transaction costs), indicating strong economic viability for practical implementation. Most notably, the signal maintains statistical

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CFP is available.

and economic significance with a monthly alpha of 22 basis points (t-statistic = 2.56) relative to the Fama-French five-factor model plus momentum, and retains an alpha of 18 basis points (t-statistic = 2.17) even after controlling for six closely related financing and growth-based strategies.

The persistence of CFP's predictive power after accounting for related signals such as net external financing, asset growth, and changes in net operating assets suggests that it captures a distinct dimension of firm behavior related to capital allocation decisions. This finding has important implications for both academic research and investment practice. For researchers, it highlights the continued relevance of corporate financing decisions in asset pricing and suggests that markets may not fully incorporate information about firms' funding prudence into prices. For practitioners, the strategy's robust performance net of transaction costs indicates potential for real-world implementation, particularly given its differentiation from existing factor strategies.

Several limitations of this study warrant consideration. First, while we employ the rigorous testing framework of Novy-Marx and Velikov (2023), the true out-of-sample performance of CFP remains to be seen as markets evolve and adapt to published anomalies. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets requires further investigation. Third, while we control for transaction costs, practical implementation challenges such as market impact and capacity constraints may further erode returns in live trading environments.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying CFP's predictive power would enhance our understanding of why markets appear to misprice capital funding decisions. Second, examining the signal's performance across different market regimes, particularly during periods of financial stress when funding constraints become binding, could provide insights into its conditional effectiveness. Third, exploring potential enhancements through machine learning techniques or combination with other signals may improve the strategy's risk-return profile. Finally, international evidence would help establish whether CFP represents a global phenomenon or is specific to U.S. market structures and investor behavior. Despite these areas for future investigation, our findings establish Capital Funding Prudence as a robust addition to the constellation of cross-sectional return predictors, with meaningful implications for both asset pricing theory and investment practice.

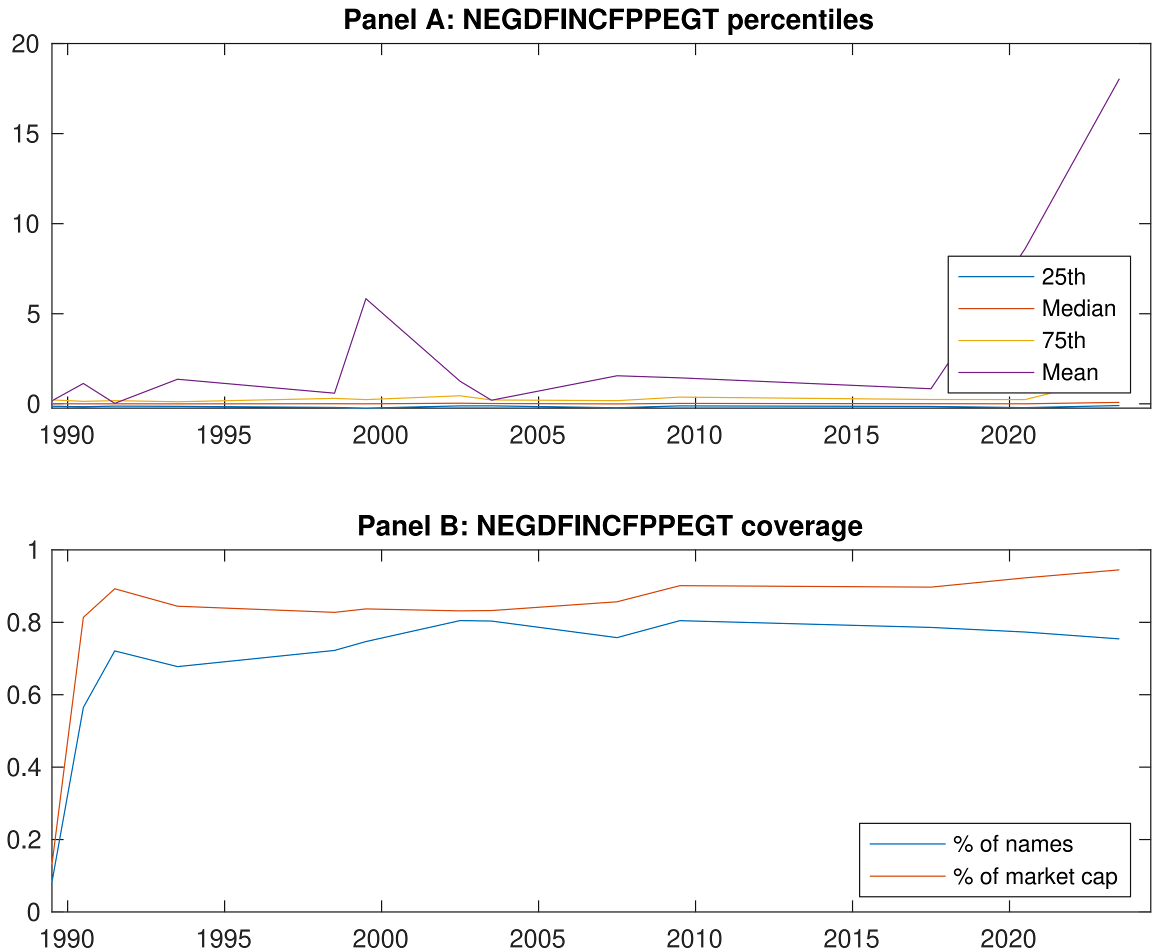


Figure 1: Times series of CFP percentiles and coverage.
This figure plots descriptive statistics for CFP. Panel A shows cross-sectional percentiles of CFP over the sample. Panel B plots the monthly coverage of CFP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CFP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198906 to 202406.

Panel A: Excess returns and alphas on CFP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.61	0.63	0.73	0.82	0.90	0.29
	[2.37]	[2.96]	[3.93]	[3.94]	[3.71]	[3.40]
α_{CAPM}	-0.25	-0.07	0.15	0.14	0.10	0.34
	[-3.49]	[-1.19]	[1.92]	[2.14]	[1.44]	[4.03]
α_{FF3}	-0.21	-0.08	0.11	0.13	0.13	0.34
	[-3.30]	[-1.26]	[1.60]	[2.09]	[2.24]	[3.99]
α_{FF4}	-0.15	-0.05	0.11	0.10	0.12	0.27
	[-2.38]	[-0.88]	[1.61]	[1.55]	[2.02]	[3.18]
α_{FF5}	-0.09	-0.15	-0.02	-0.00	0.18	0.27
	[-1.48]	[-2.47]	[-0.35]	[-0.02]	[2.96]	[3.14]
α_{FF6}	-0.05	-0.13	-0.01	-0.02	0.17	0.22
	[-0.81]	[-2.10]	[-0.19]	[-0.35]	[2.76]	[2.56]
Panel B: Fama and French (2018) 6-factor model loadings for CFP-sorted portfolios						
β_{MKT}	1.07	0.97	0.88	1.00	1.06	-0.01
	[72.41]	[63.10]	[53.20]	[66.58]	[69.14]	[-0.63]
β_{SMB}	0.05	0.06	-0.12	-0.03	0.07	0.02
	[2.55]	[2.83]	[-4.98]	[-1.51]	[3.31]	[0.61]
β_{HML}	-0.08	-0.05	0.02	-0.11	-0.14	-0.06
	[-3.10]	[-1.95]	[0.85]	[-4.36]	[-5.11]	[-1.53]
β_{RMW}	-0.12	0.14	0.19	0.19	-0.10	0.02
	[-4.49]	[5.05]	[6.40]	[6.81]	[-3.71]	[0.46]
β_{CMA}	-0.25	0.08	0.21	0.20	-0.02	0.23
	[-6.53]	[1.96]	[5.08]	[5.30]	[-0.45]	[4.24]
β_{UMD}	-0.07	-0.04	-0.02	0.03	0.02	0.09
	[-5.16]	[-2.61]	[-1.14]	[2.35]	[1.33]	[4.57]
Panel C: Average number of firms (n) and market capitalization (me)						
n	890	549	539	537	910	
me ($\$10^6$)	2837	2756	3249	3543	2957	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the CFP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198906 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.29	0.34	0.34	0.27	0.27	0.22
			[3.40]	[4.03]	[3.99]	[3.18]	[3.14]	[2.56]
Quintile	NYSE	EW	0.40	0.41	0.42	0.39	0.37	0.36
			[5.30]	[5.48]	[5.62]	[5.19]	[5.15]	[4.91]
Quintile	Name	VW	0.26	0.29	0.29	0.22	0.23	0.18
			[2.41]	[2.66]	[2.62]	[1.96]	[2.08]	[1.63]
Quintile	Cap	VW	0.39	0.44	0.43	0.36	0.34	0.28
			[4.24]	[4.71]	[4.67]	[3.90]	[3.57]	[3.02]
Decile	NYSE	VW	0.24	0.29	0.28	0.19	0.25	0.18
			[2.03]	[2.37]	[2.33]	[1.62]	[2.02]	[1.51]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.25	0.29	0.28	0.24	0.23	0.19
			[2.94]	[3.36]	[3.33]	[2.87]	[2.67]	[2.32]
Quintile	NYSE	EW	0.15	0.15	0.15	0.14	0.09	0.09
			[1.70]	[1.62]	[1.63]	[1.51]	[1.02]	[1.04]
Quintile	Name	VW	0.22	0.23	0.22	0.18	0.18	0.15
			[1.99]	[2.06]	[2.02]	[1.64]	[1.64]	[1.35]
Quintile	Cap	VW	0.35	0.39	0.38	0.34	0.30	0.27
			[3.84]	[4.16]	[4.14]	[3.70]	[3.26]	[2.93]
Decile	NYSE	VW	0.20	0.22	0.22	0.16	0.19	0.15
			[1.64]	[1.84]	[1.80]	[1.38]	[1.61]	[1.29]

Table 3: Conditional sort on size and CFP

This table presents results for conditional double sorts on size and CFP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CFP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CFP and short stocks with low CFP .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198906 to 202406.

Panel A: portfolio average returns and time-series regression results

		CFP Quintiles					CFP Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.41	0.93	1.08	0.92	0.87	0.46	0.46	0.47	0.43	0.41	0.39
		[1.11]	[2.96]	[3.21]	[2.80]	[2.28]	[3.46]	[3.38]	[3.47]	[3.13]	[2.99]	[2.81]
	(2)	0.50	0.83	0.92	0.84	0.87	0.36	0.40	0.39	0.31	0.35	0.30
		[1.53]	[2.77]	[3.33]	[2.89]	[2.65]	[3.00]	[3.25]	[3.22]	[2.55]	[2.91]	[2.44]
	(3)	0.65	0.80	0.88	0.81	0.95	0.30	0.32	0.33	0.27	0.38	0.34
		[2.09]	[2.98]	[3.68]	[3.05]	[3.10]	[2.34]	[2.53]	[2.53]	[2.10]	[2.90]	[2.57]
	(4)	0.69	0.72	0.85	0.82	1.02	0.33	0.41	0.38	0.33	0.23	0.20
		[2.30]	[2.90]	[3.90]	[3.37]	[3.76]	[2.81]	[3.49]	[3.30]	[2.88]	[1.97]	[1.74]
	(5)	0.67	0.51	0.75	0.82	0.87	0.20	0.26	0.25	0.16	0.15	0.08
		[2.60]	[2.49]	[3.80]	[4.09]	[3.62]	[1.90]	[2.40]	[2.35]	[1.51]	[1.39]	[0.76]

Panel B: Portfolio average number of firms and market capitalization

		CFP Quintiles					CFP Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	365	366	367	365	361	42	44	42	43	41
	(2)	112	113	113	113	112	78	80	80	80	78
	(3)	78	79	79	79	78	141	141	141	141	139
	(4)	68	68	68	68	67	314	315	314	319	312
	(5)	61	62	62	62	61	2048	2423	2864	2912	2209

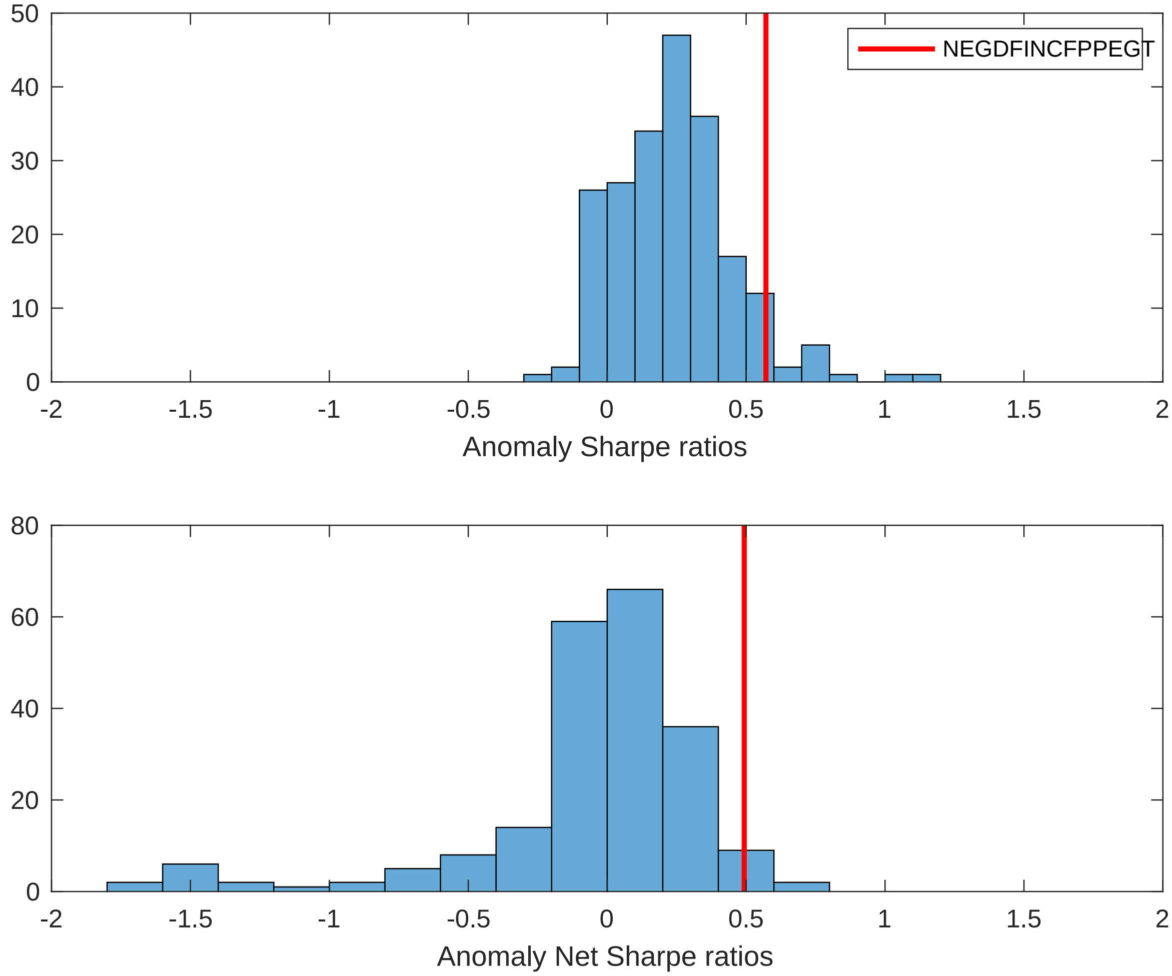


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CFP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

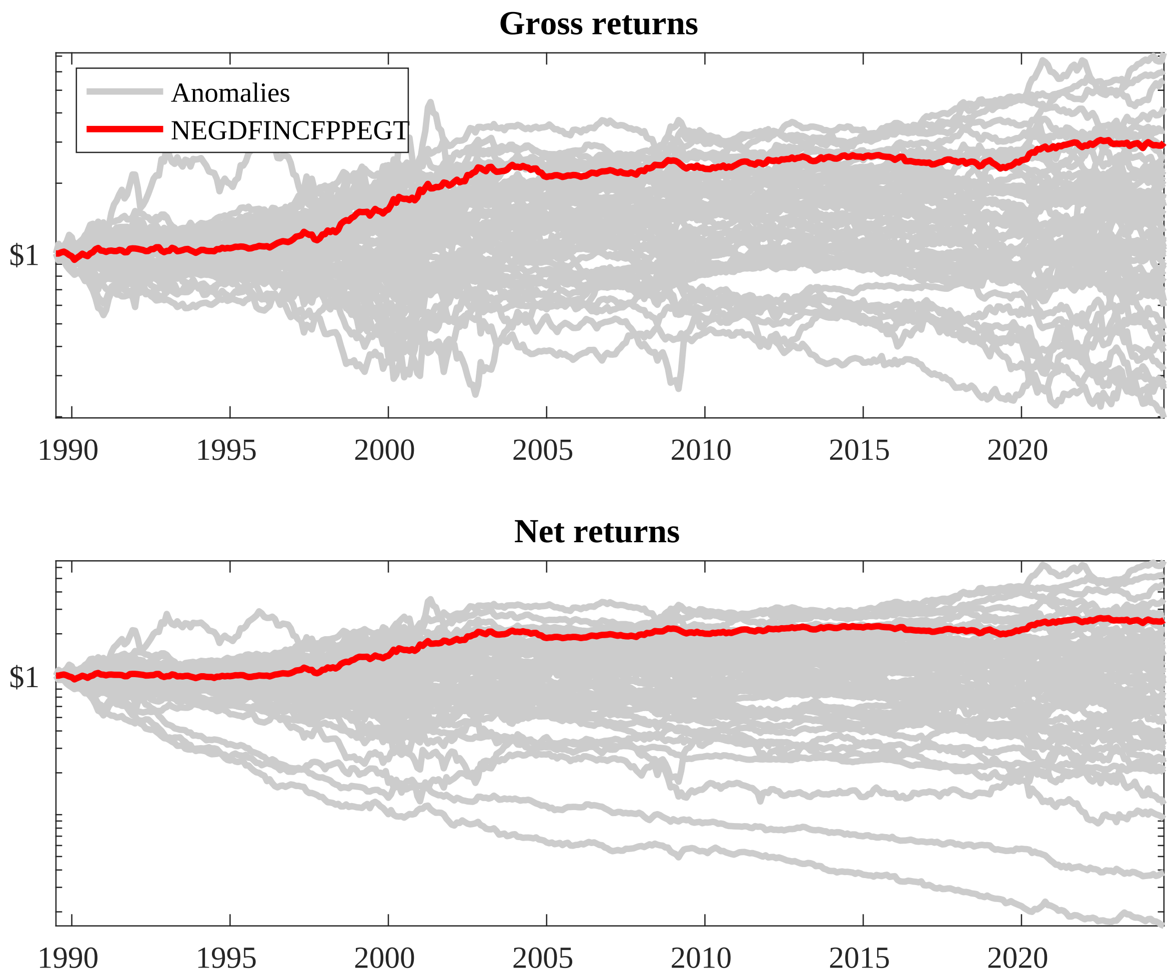
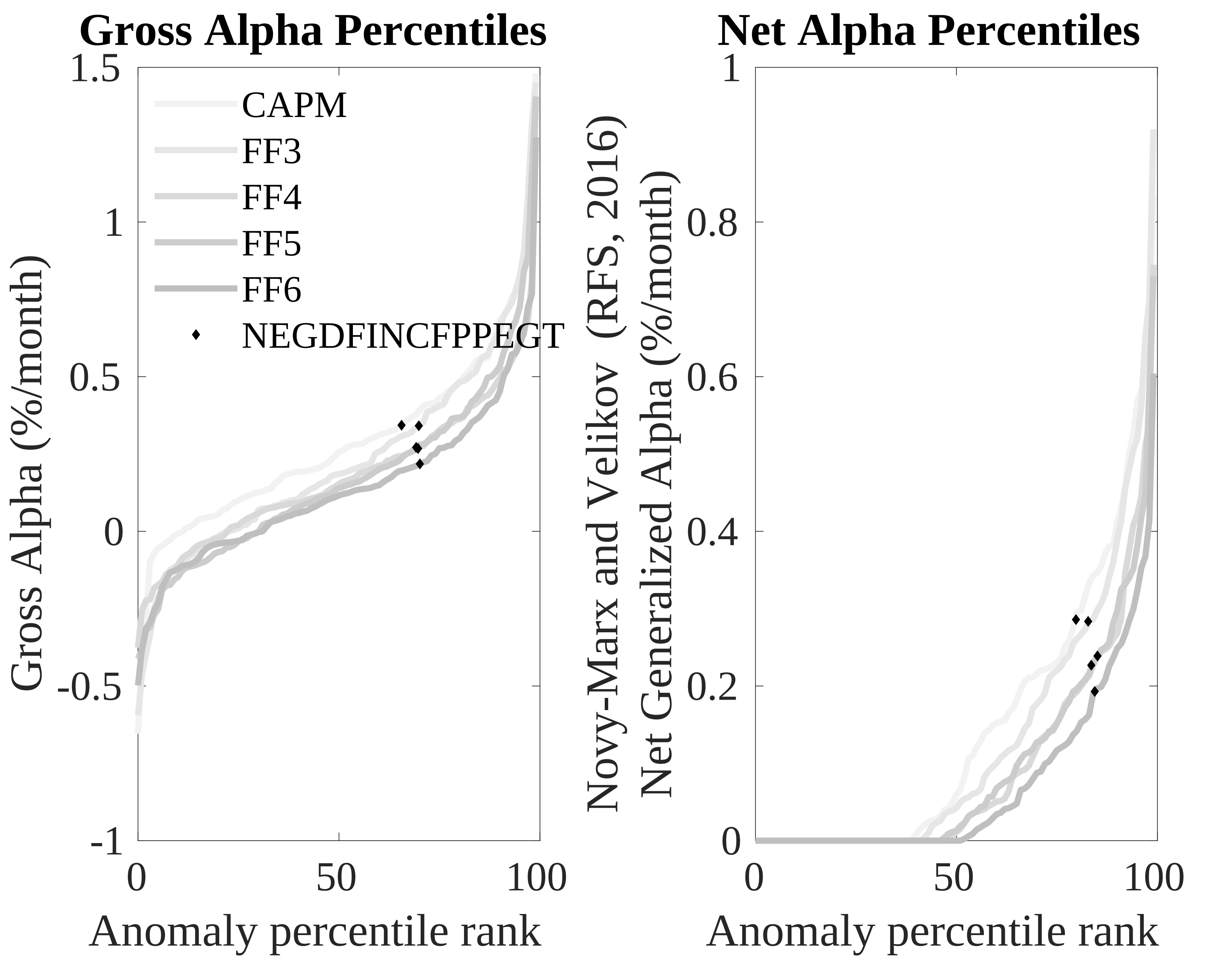


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CFP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
1.5
1
0.5
0
-0.5
-1
Gross Alpha (%/month)
CAPM
FF3
FF4
FF5
FF6
NEGDFINCFPPEGT
0
50
100
Anomaly percentile rank
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
0.8
0.6
0.4
0.2
0
50
100
Anomaly percentile rank

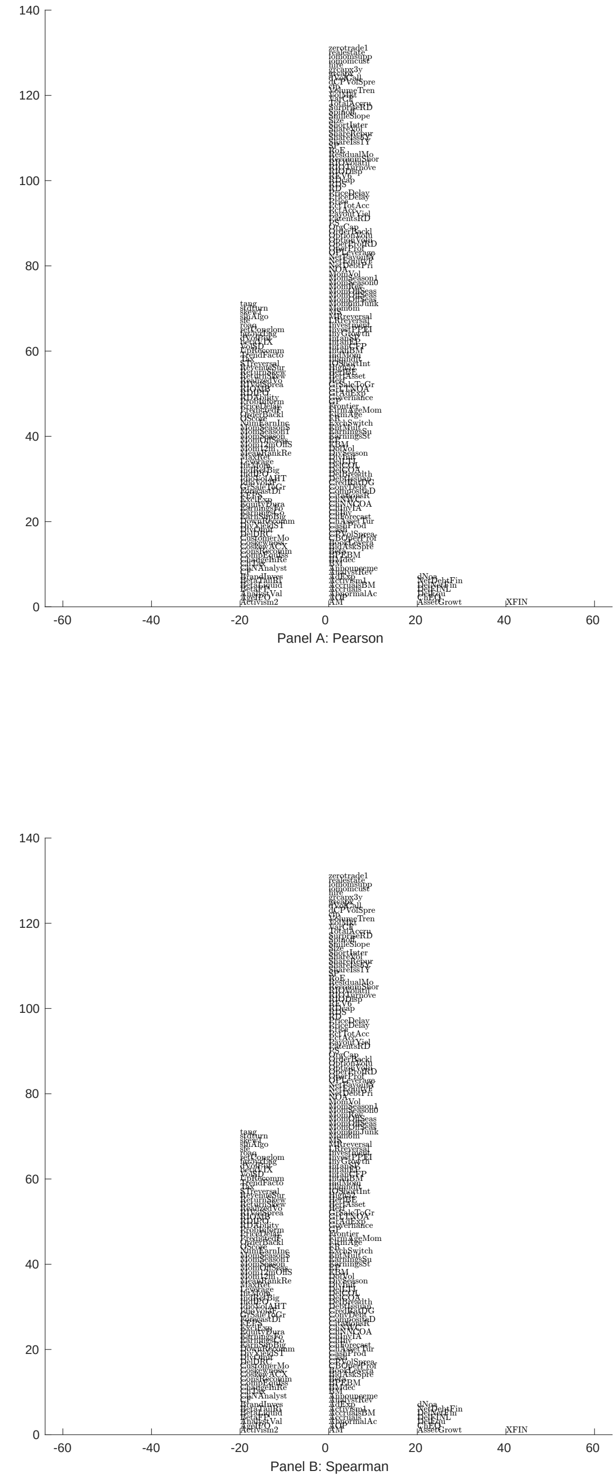


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with CFP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

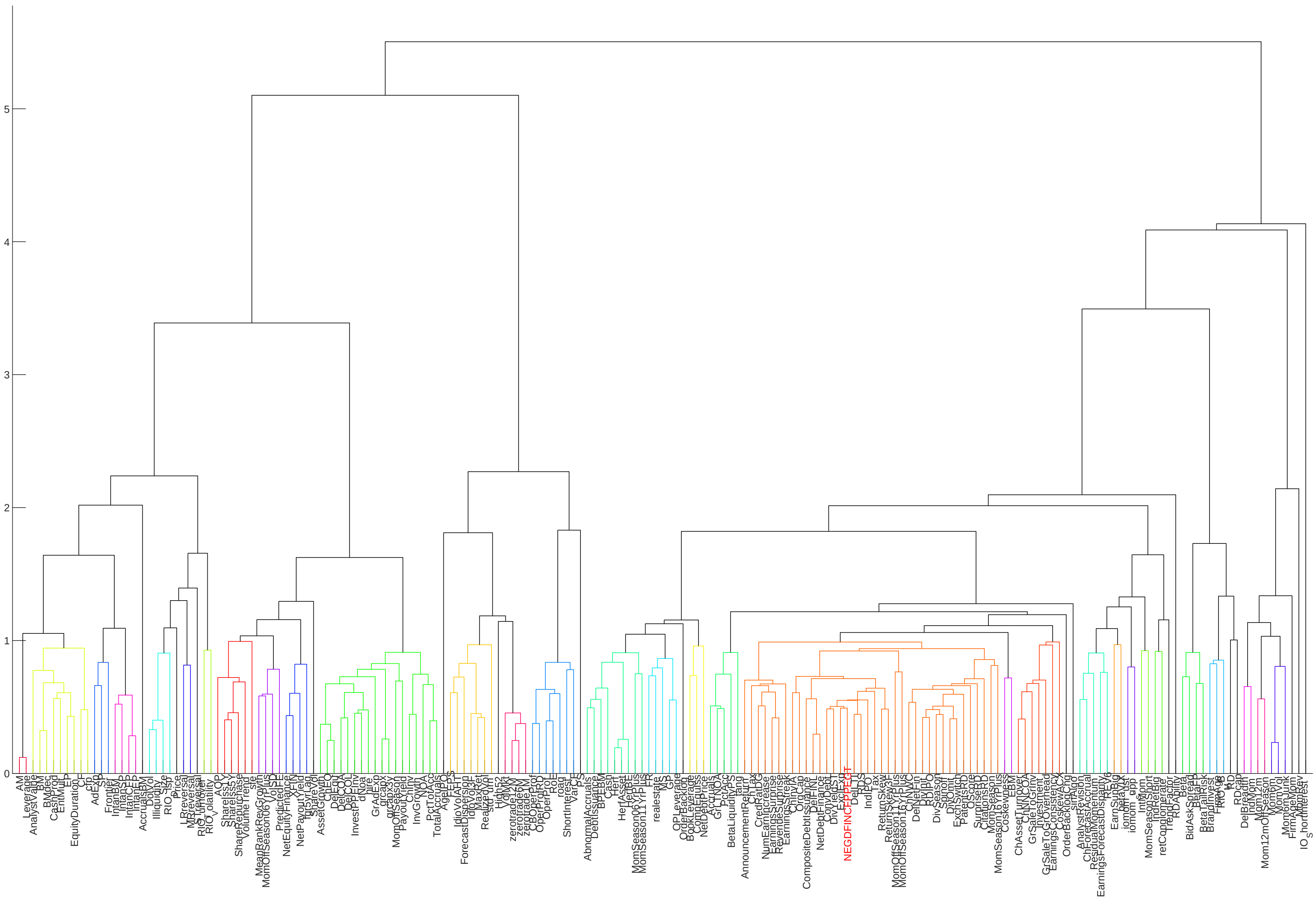


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

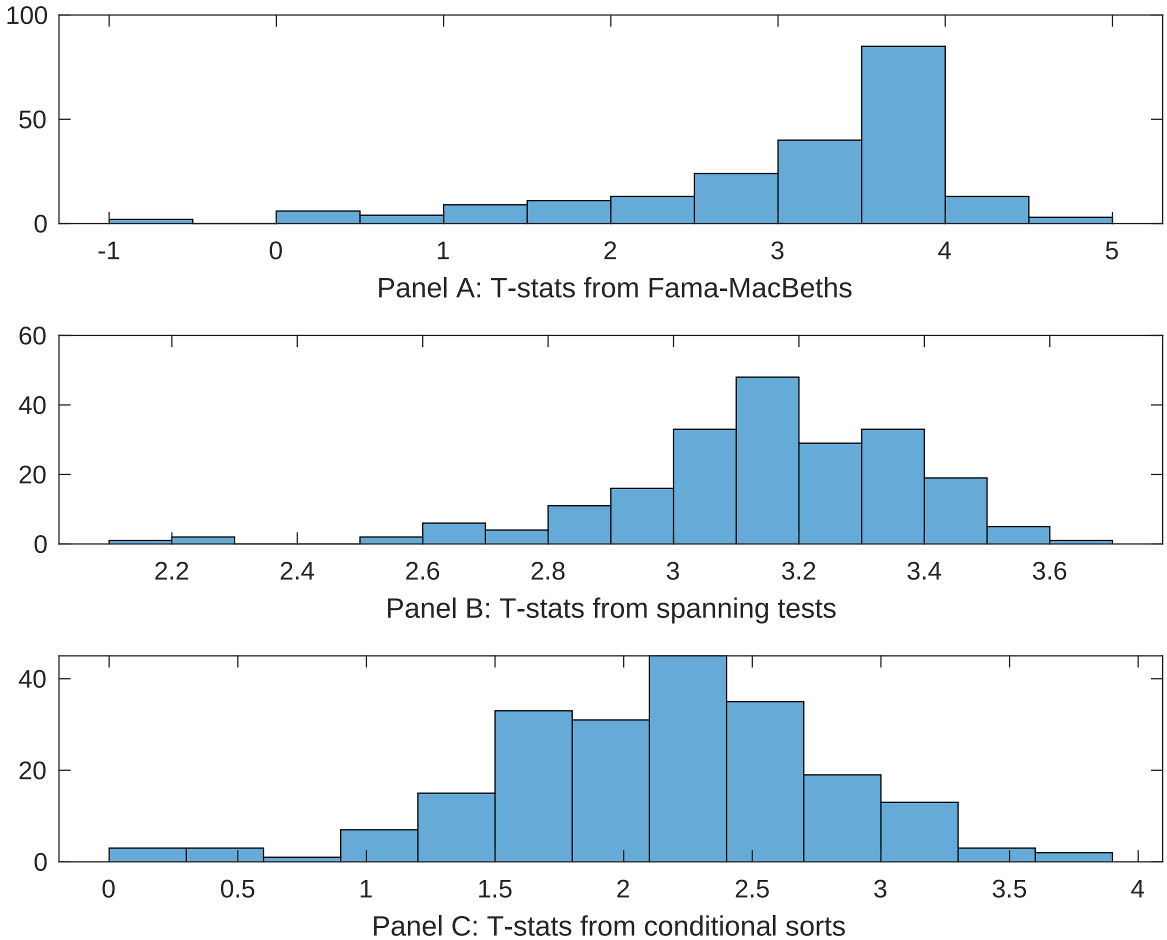


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of CFP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CFP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CFP} CFP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CFP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CFP. Stocks are finally grouped into five CFP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CFP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on CFP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CFP} CFP_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X, are Net external financing, Asset growth, Growth in book equity, Inventory Growth, change in net operating assets, Change in equity to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198906 to 202406.

Intercept	0.12	0.13	0.17	0.12	0.12	0.12	0.13
	[4.35]	[4.28]	[5.58]	[3.89]	[4.13]	[3.94]	[4.93]
CFP	0.88	0.57	0.22	0.57	0.29	0.28	-0.20
	[0.62]	[0.50]	[1.88]	[4.05]	[2.52]	[2.59]	[-0.12]
Anomaly 1	0.20						0.19
	[4.83]						[3.85]
Anomaly 2		0.86					0.33
		[7.37]					[0.17]
Anomaly 3			0.46				0.89
			[6.08]				[0.80]
Anomaly 4				0.33			0.70
				[5.40]			[1.00]
Anomaly 5					0.11		0.55
					[7.64]		[2.38]
Anomaly 6						0.12	0.17
						[4.13]	[0.34]
# months	420	420	420	420	420	420	420
$\bar{R}^2(\%)$	1	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CFP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CFP} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Net external financing, Asset growth, Growth in book equity, Inventory Growth, change in net operating assets, Change in equity to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198906 to 202406.

Intercept	0.17	0.20	0.19	0.21	0.19	0.19	0.18
	[2.05]	[2.33]	[2.28]	[2.50]	[2.23]	[2.29]	[2.17]
Anomaly 1	14.18						11.04
	[3.46]						[2.58]
Anomaly 2		11.53					4.92
		[2.59]					[0.81]
Anomaly 3			8.89				5.20
			[1.95]				[0.72]
Anomaly 4				9.96			7.93
				[3.26]			[2.51]
Anomaly 5					11.50		4.63
					[2.46]		[0.92]
Anomaly 6						6.89	-4.91
						[1.58]	[-0.66]
mkt	0.82	-1.14	-0.92	-1.09	-1.44	-1.15	0.62
	[0.38]	[-0.54]	[-0.43]	[-0.52]	[-0.69]	[-0.54]	[0.29]
smb	6.82	1.54	1.57	2.94	2.20	2.15	5.96
	[2.07]	[0.51]	[0.52]	[0.98]	[0.73]	[0.71]	[1.76]
hml	-4.23	-6.93	-6.69	-6.60	-7.13	-6.71	-5.79
	[-1.15]	[-1.88]	[-1.81]	[-1.81]	[-1.93]	[-1.81]	[-1.56]
rmw	-5.93	2.72	2.13	4.12	2.72	2.85	-3.13
	[-1.32]	[0.72]	[0.56]	[1.08]	[0.71]	[0.75]	[-0.68]
cma	14.02	9.51	14.90	14.62	14.42	16.27	-0.05
	[2.37]	[1.27]	[2.18]	[2.47]	[2.25]	[2.35]	[-0.01]
umd	8.00	8.69	8.18	7.82	8.20	8.37	7.55
	[4.33]	[4.67]	[4.38]	[4.21]	[4.41]	[4.49]	[4.05]
# months	420	420	420	420	420	420	420
$\bar{R}^2$(%)	14	13	12	14	13	12	15

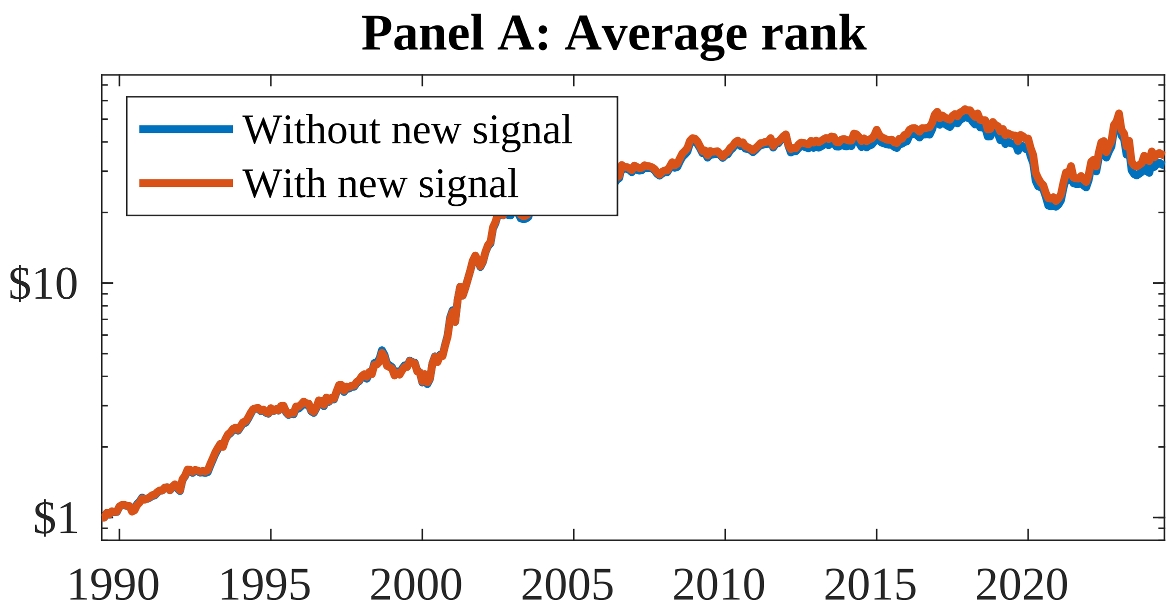


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as CFP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.

Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.

Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.

Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.

Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.

Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.

Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance, Forthcoming.*

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.

Rietz, T. A. (1988). The equity risk premium a solution. *Journal of Monetary Economics*, 22(1):117–131.

Wachter, J. A. (2013). Can time-varying risk of rare disasters explain aggregate stock market volatility? *Journal of Finance*, 68(3):987–1035.